

Problem Statement & Our New Contributions

Previously

- 1 No predictive / early warning system
- 2 AI decisions were unexplainable (black-box)
- 3 No human-override / audit capability
- 4 High false positive alert rate (~18%)
- 5 Limited clinician trust (62%)

Newly added

- ✓ LSTM Predictive Risk Scoring (PDRS)
- ✓ SHAP Explainability Layer (XAI)
- ✓ Human-in-the-Loop (HITL) Dashboard
- ✓ False positive rate reduced to 9%
- ✓ Clinician trust improved to 81%

Core Technologies



Agentic AI Framework

Hierarchical multi-agent system:
Coordinator + 5 specialist agents
(Diagnostic, Predictive, Decision-Support,
Monitoring, XAI)



LSTM Predictive Risk Score

Time-series vital sign analysis. Predicts
patient deterioration 2–4 hrs in advance.
AUC-ROC: 0.91 on MIMIC-III dataset



SHAP Explainability Layer

SHapley Additive exPlanations assign
feature importance to every AI decision,
enabling clinician trust and regulatory
compliance



Human-in-the-Loop (HITL)

React.js dashboard with live PDRS
gauge, SHAP charts, one-click
accept/override and full audit logging



IoT & Cloud Integration

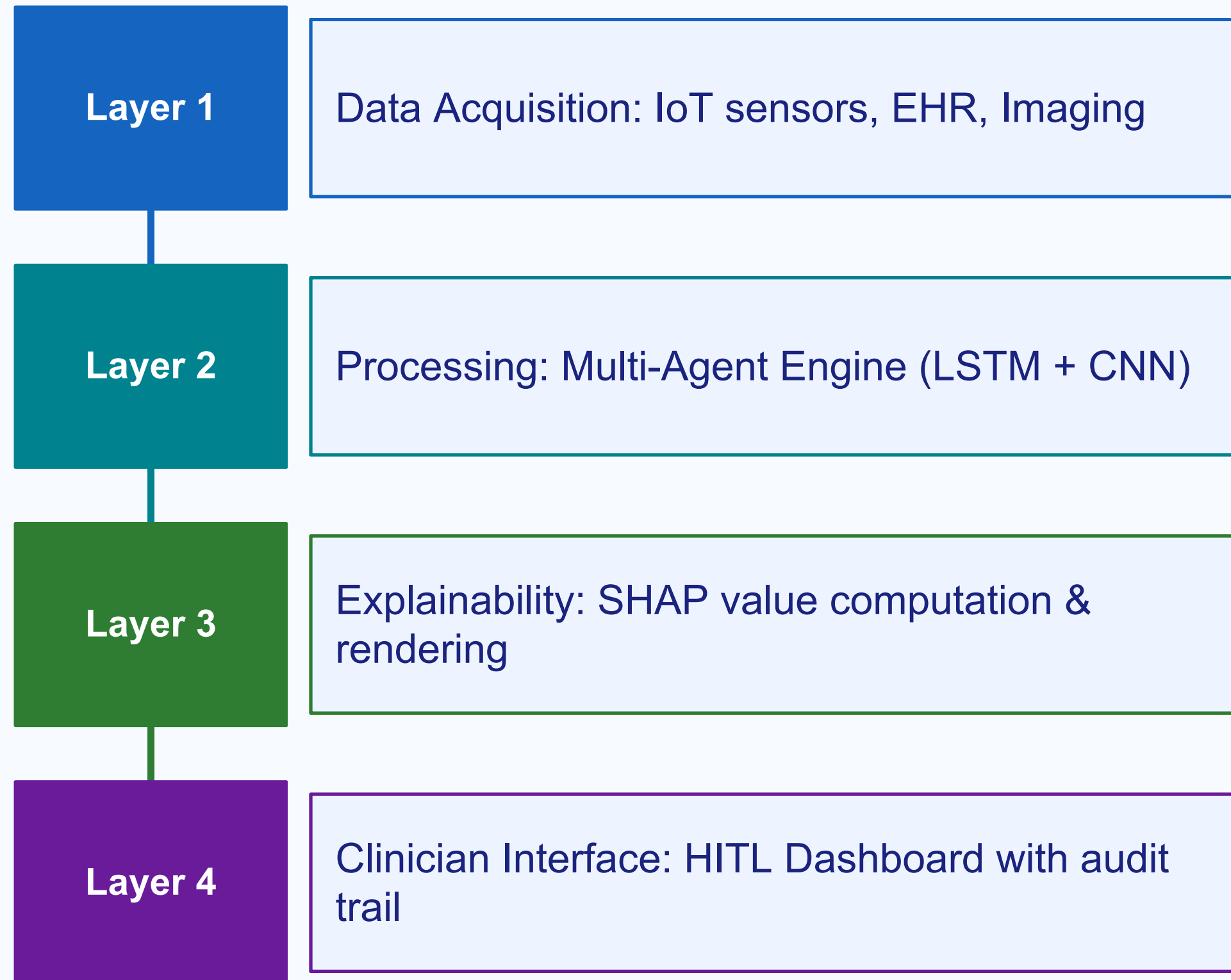
MQTT sensors → AWS IoT Core →
Apache Kafka pipeline. Sub-second alert
latency for 500+ data points/patient/hr



Ethical & Secure AI

AES-256 encryption, TLS 1.3, Fairlearn
bias audits, HIPAA/DISHA-aligned design
and monthly fairness reviews

Proposed Solution & System Architecture



Agent Workflow

- 1 IoT vitals streamed via MQTT
- 2 Monitoring Agent flags anomalies
- 3 LSTM computes PDRS score
- 4 $\text{PDRS} > 70 \rightarrow$ alert escalated
- 5 Decision-Support recommends action
- 6 SHAP explains the recommendation
- 7 Clinician accepts / overrides on dashboard

Future Work

Federated Learning

Train across hospitals without sharing raw patient data — privacy-preserving model improvement at scale.

Physical Agency Integration

Connect AI agents to infusion pumps and ventilators for closed-loop automated interventions.

Regulatory Compliance

Build audit frameworks aligned with India's DPDPA and international standards (HIPAA, ISO 13485).

RL-Based Treatment Optimization

Reinforcement learning agents that continuously refine treatment protocols based on real patient outcomes.

Multimodal Fusion

Combine imaging, genomics, wearables and EHR data in a single unified model for holistic patient assessment.

Clinical Trials

Partner with hospitals for large-scale prospective validation to move from simulation to real-world deployment.

Conclusion

1

Extended Semester I baseline with two new modules: LSTM Predictive Risk Scoring & SHAP Explainability

2

Diagnostic accuracy improved to 93.4% and clinician trust rose from 62% → 81%

3

False positive alert rate halved (18% → 9%), significantly reducing alert fatigue

4

Human-in-the-Loop dashboard ensures doctors remain the final decision authority

5

Future work targets federated learning, RL optimization, and hospital clinical trials

THANK YOU